Urban noise modelling in Hanoi

End-of-internship progress report

Laurian Jamin and Lucas Zborowski

ISIMA, Clermont-Ferrand, France
Research interns, COSMOS Lab, VinUniversity, Hanoi

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This report is submitted jointly by both authors, who carried out the work together and share responsibility for all of it.

Supervised by Doanh Nguyen-Ngoc, VinUniversity.
With Nguyen Thanh Quang, VinUniversity, who will run the distribution of the field application inside VinUniversity (§6.1).

Every number in this report is quoted from an artefact in the repository and the artefact is named where the number appears. No figure here was drawn for this report: all come from the project's own generating scripts (Appendix A), and the application screens are drawn from its Compose source.

Contents

1 Context, problem and objectives

1.1 The problem

Hanoi is dense, motorcycle-dominated and growing quickly, and it has no routine environmental-noise monitoring programme and no statutory noise action plan (Nguyen et al., 2025a). Class 1 and class 2 sound level meters and commercial propagation software are out of reach of most local research budgets. The practical question this internship attacks is therefore not "what is the noise map of Hanoi" but:

> **How far does a noise-mapping protocol built from three consumer smartphones and open data actually get you, and where, exactly, does it break?**

1.2 The plan as it started, and the plan as it ended

The project began in June 2026 from the *Sunbird Urban Noise Uganda 61K* dataset and its *Scientific Data* descriptor (Nsumba et al., 2026). The intended chain was: reproduce the published figures, train a morphology-to-level surrogate on Uganda, transfer it to Hanoi, and use a local field campaign only to calibrate an offset (`docs/project-timeline.md`, June 2026).

Two things changed that. Transfer failed outright, and an internal scientific audit on 5 August 2026 (`docs/audit/scientific-audit.md`) found three cumulative problems in the chain: metrology with no absolute anchor, a cross-validation that leaked and masked a negative leave-one-site-out score, and a published map whose effective resolution was incompatible with what it claimed to represent. A previously advertised $R^2 = 0.45$ was withdrawn.

The project therefore **pivoted to a methodological study**: no professional sound level meter would become available, the field campaign was closed for good, and the object of study became the protocol itself. The negative results became the contribution.

1.3 Research questions, as they now stand

- **RQ1.** Can urban morphology from OpenStreetMap predict measured noise at an *unmeasured* location in Hanoi?
- **RQ2.** Does a model trained in one city *transfer* to another?
- **RQ3.** Does vehicle *flow extracted from video* explain measured levels?
- **RQ4.** What can a smartphone protocol claim, given that it has *no absolute reference*?

Explicitly out of scope: night-time noise (10 of 363 measurements fall outside the 06:00–21:00 window), compliance assessment against any standard, and any prediction outside the three measured sites.

2 Background and related work

This section lists only what the repository actually uses. The project's own selection rule is *actual use*: a reference is cited because it informed a decision, a parameter or a stated limitation (`docs/literature-review.md`). Seventeen entries are held in `docs/references.bib`; every DOI was resolved against the Crossref API on 12 August 2026.

Source dataset and method being reproduced. The Sunbird Urban Noise Uganda 61K dataset (Nsumba et al., 2026) supplies both the pretraining corpus for the transfer experiment and the morphology-to-level methodology this work reproduces.

Instrumented Vietnamese campaigns, used as absolute anchors. Because the campaign has no reference instrument, the level distribution is anchored against published campaigns that did have one: Phan et al. (2010) (Hanoi, RION NL-21/NL-22, 24 h continuous, seven sites) and Gelb and Apparicio (2019) (Ho Chi Minh City, personal dosimeters with GPS over 3 300 segments). Horn behaviour, a first-order noise source in Vietnamese traffic, is characterised by Nguyen et al. (2025b); the regulatory vacuum by Nguyen et al. (2025a).

Standards. QCVN 26:2010/BTNMT (Ministry of Natural Resources and Environment of Viet Nam, 2010) supplies the only thresholds this project displays (70 dB day / 55 dB night) and the 06:00–21:00 day boundary. ISO 1996-2 (International Organization for Standardization, 2017) was read and is used mainly as a list of what the protocol does *not* control: wind speed, microphone height, distance to reflecting surfaces, integration duration.

Smartphone metrology. Consumer smartphone sound measurement is known to depart from reference instruments by several decibels in a handset-, OS- and level-dependent way that is not a constant offset (Kardous and Shaw, 2014; Murphy and King, 2016). This is the load-bearing citation behind the project's central limitation.

Spatial validation. Roberts et al. (2017) is the reference behind the withdrawal of the $R^2 = 0.45$: a cross-validation grouped on cells smaller than the feature aggregation radius leaks between folds. Meyer and Pebesma (2021) is the area-of-applicability argument behind the rule that no map is published outside the sampled envelope.

Physical propagation, the indicated continuation. CNOSSOS-EU (Kephalopoulos et al., 2014) and its open implementation NoiseModelling (Bocher et al., 2019) are cited as the propagation kernel the results argue for, and are not implemented here.

One known gap in the bibliography

The twelve journal quartiles are recorded as unread rather than filled in: `scimagojr.com` returns HTTP 403 to automated requests, and a search engine's summary of a quartile is second-hand evidence, which this project classifies as grey everywhere else. The Phan et al. (2010) L_{den} *values used as an anchor are taken from secondary sources and flagged* `to_check` *in* `scripts/06_anchor_literature.py`.

3 Work done

3.1 Chronology

Reconstructed from `git log` (150 commits, 1 June to 21 August 2026).

Period	What happened
1–12 June	Repository opened; Sunbird pipeline reproduced; large surrogate and convention-invariant v2 features trained; Barcelona diagnostic; field forms v2 and the construction register written; GAMA plan drafted.
10 June – 22 July	**Field campaign.** Carried mainly by the two interns, with occasional support from a research assistant; three consumer handsets, ODK Collect against a KoboToolbox server, three contrasted typologies. Pipeline re-run as the sample grew, from 107 points to 220. Transfer abandoned in favour of direct local training on 30 June.
15–31 July	Repository restructured, YOLO vehicle counting, weather analysis, seven-page report; first deck and GAMA model. This is the version that carried the $R^2 = 0.45$.
5 August	**Audit and pivot.** Nine corrections applied the same day; V2 built: physical kernel, object tracking, dashboard, updated GAMA.
6 August	Measured flow restricted to a 150 m corridor around measurement points.
11–12 August	**Restructuring for handover:** target layout, installable `noise_hanoi` package, Makefile and first tests, pickled Uganda models replaced by text boosters, licences and `CITATION.cff`, handover, methodology, full French→English pass across scripts, notebooks, GAMA and docs.
20–21 August	**Android field application** (and the `feat/mobile-app` branch); Uganda transfer made reproducible; presentation brief; report typeset in LaTeX; presentation figures, decks and speaker scripts (. . .).

3.2 Field campaign and protocol

363 measurements were taken between 10 June and 22 July 2026 across three deliberately contrasted urban typologies, together with 147 timestamped traffic videos (6.0 GB, not distributed). Collection used ODK Collect against a KoboToolbox server with two XLSForms: a 16-question noise form and a 6-question construction register (`data/forms/`).

The protocol (`docs/field-protocol.md`) fixes a 20–30 s stationary observation, a 10 m GPS accuracy gate, and a recorded outdoor distance to the road. **The three handsets were cross-calibrated against one another**: placed side by side, read simultaneously, trimmed to agree within about 1 dB, middle device taken as reference, and against no standard. No reference instrument was available and none could be borrowed.

3.3 Metrological framing

That decision governs everything downstream, and the project states its consequence rather than working around it (`docs/metrology.md`). The recorded quantity is denoted $L_{A,25\mathrm{s}}$, a single A-weighted, SLOW-weighted level over a 20–30 s stationary observation, and never simply “dB”, and never `L_Aeq`, `L_den` or `L_night`.

- **Differences are supported**: between places, hours and typologies, because a bias common to the three devices cancels in a difference.
- **Absolute levels are not certified.** They are reported with a bounded bias interval (+3.0 to +7.3 dB, `results/report/numbers.tex`), estimated by anchoring against instrumented literature and *never applied* to the data.
- **A WHO $L_{\mathrm{den}}/L_{\mathrm{night}}$ comparison present in earlier versions was withdrawn in full**: it compared a 25 s sample to an annual long-term indicator.
- Exceedances are reported as “the proportion of short-sample measurements exceeding the QCVN daytime threshold value”, a descriptive statistic, and never as a rate of regulatory non-compliance.

3.4 The processing chain

The pipeline is twelve numbered scripts in `scripts/`, executed in order, with shared code in the installable package `src/noise_hanoi/`:

Step	Script	What it does
01	`01_prepare_field_data`	Clean the Kobo export, join hourly weather
02	`02_count_vehicles`	YOLOv8 + ByteTrack, line-crossing flow per class
02b	`02b_fetch_osm_extract`	Fetch the OSM extract step 03 requires
03	`03_build_features`	Morphology within a 300 m radius
04	`04_evaluate_models`	12 models × 3 CV protocols → `metrics.json`
05	`05_calibrate_emissions`	Non-negative energy regression on vehicle counts
06	`06_anchor_literature`	Bias interval against instrumented campaigns
07	`07_export_gama_inputs`	40 m grid × 17 hours, GIS layers for GAMA
08	`08_validate_simulation`	Simulated vs measured levels
09/09b	`09_build_field_map`, `09b_build_analyses`	Field map, five analyses, exceedances
10/11	`10_build_report`, `11_build_dashboard`	Typeset report, static dashboard
12	`12_presentation_figures`	The seven vector figures used in this report

3.5 The evaluation framework

This is the part of the work we are most confident in. Every model is scored under three splits of increasing severity, on *identical folds*:

1. **600 m spatial block cross-validation**, 17 blocks, the permissive protocol;
2. **buffered leave-one-out with a 300 m exclusion**, the *reference* protocol, because its exclusion radius equals the feature aggregation radius;
3. **leave-one-site-out**, each typology predicted from the other two.

95 % confidence intervals come from a 2 000-replicate block bootstrap. Eight models are compared against baselines that include a lookup table of mean level by (site, hour), which uses no spatial variable at all, and against ablations that isolate morphology from time.

The delivered model is chosen by code, not by hand: `04_evaluate_models.py` takes the best score under the reference protocol among candidates fixed in advance, writes the choice into `models/metrics.json`, and the export reads a flag. This was a direct response to the audit: the published map must not be able to inherit, silently, a model that only wins on a permissive split.

3.6 The agent-based simulation

`simulation/gama/hanoi_noise.gaml` reads the exported 40 m grid and adds **receiver agents**. The design decision is recorded and argued (`simulation/gama/README.md`): emitter agents, motorcycles, cars, would need real per-street flows, fleet composition, speeds and signal cycles, none of which exist for Hanoi, so such a simulation would be neither calibratable nor verifiable. Receiver agents move through the area and accumulate a *noise dose* from a map that is measured and validated; nothing is invented.

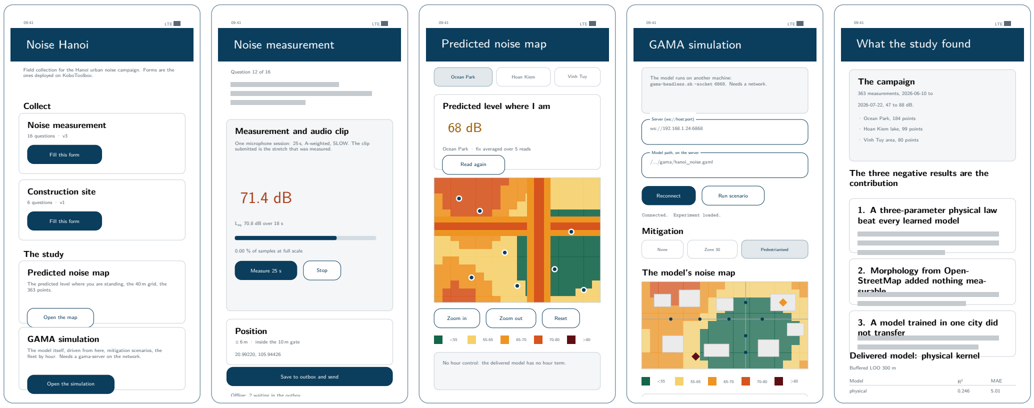
Tier 1 (interactive map, hour and traffic sliders, live share-above-threshold indicator) is implemented. Tier 2 (exposed pedestrian agents with daily journeys) is **not implemented** and remains the most valuable extension. Tier 3 (scenarios) is partial.

One correction matters here: the traffic multiplier scales the *traffic share* of the energy and never the background. Applied to total energy it double-counts the background and inflates the apparent benefit of mitigation, the claimed gain from pedestrianisation fell from 7.0 dB to 3.5 dB once it was applied correctly.

3.7 The Android field application

Built 20–21 August (`mobile/`, Kotlin + Jetpack Compose, 6 679 lines across 36 `.kt` files, 36 unit tests). It replaces the ODK Collect + Decibel X pairing with one offline-first application: the two forms, a 25 s A-weighted SLOW measurement from `AudioRecord` with AGC, noise suppression and echo cancellation switched off, the audio clip from the same microphone session, the 10 m GPS gate, an outbox drained by `WorkManager`, and OpenRosa submission to KoboToolbox.

It also carries the study: the published 40 m grid by hour, the 363 points, the delivered three-parameter kernel evaluated where the user stands, with an explicit refusal outside the mapped areas, and the results with the negative ones stated as plainly as the positive one. Tier 1 of the GAMA model is recomputed on the phone and agrees with the model itself to within 0.2 dB from ×0.2 to ×3 traffic. The GAMA screen drives the real model over `gama-server`; end-to-end figures from the app match the model driven directly (62.97 dB unmitigated, 60.53 dB under pedestrianisation).



Home Measure Map GAMA Results

Figure 1: The five screens of the field application. These are vector reconstructions, not device captures: every string, colour and control is read from the Compose source (`mobile/app/src/main/java/org/noisehanoi/mobile/ui/`), screen by screen, `HomeScreen.kt`, `FormScreen.kt`, `MapScreen.kt`, `GamaScreen.kt`, `ResultsScreen.kt`, and the map's colour bands are `levelColour` from `Theme.kt`. Drawing them rather than screenshotting them is deliberate: nobody has yet run the application on a handset in front of a calibrated source, and a screenshot would imply otherwise. `appscreens.tex`.

The application is an instrument, not a finding

No reading from it has ever been compared to a sound level meter. Its calibration screen exists for exactly that and has never been used against a reference. It makes no compliance claim, it does not predict outside the three measured areas, and GAMA does not run on the phone.

3.8 Reproducibility and governance

Three rules were adopted after the audit and are enforced by executable checks rather than by review (`docs/handover.md` §0):

- **No metric is copied by hand.** `models/metrics.json` is the single home of every published number; the report, the dashboard, the app and the manuscript read it. `10_build_report.py` refuses to run without it.
- **No prediction outside the sampled envelope**, `tests/test_grid_extent.py` fails if the grid leaves the three sites plus 400 m.

- **The cross-validation buffer must not be smaller than the feature radius**, `tests/test_cv_protocols.py` asserts `BLOO_RADIUS_M >= FEATURE_RADIUS_M` and that a 110 m buffer would still leak.

Retracted work is archived with its reason rather than deleted, all under `docs/archive/`: the Bach Khoa map (`bach-khoa/`), the July deck (`july-deck/`) and the superseded validation (`validation-2026-08-05/`).

4 Results

4.1 The campaign

Site	n	**mean**	**median**	**min**	**max**	**>70 dB (day)**
Ocean Park (new development)	184	65.9	67.5	47.0	88.0	35.6 %
Hoan Kiem lake (old quarter)	99	69.3	70.1	55.4	80.0	50.5 %
Vinh Tuy area (transport corridor)	80	65.4	65.9	52.7	76.0	26.2 %
All	363	66.7	68.0	47.0	88.0	37.4 %

Levels in $L_{A,25s}$. Computed from `data/processed/measurements.csv`; the exceedance column is the daytime (06:00–21:00, $n = 353$) proportion above the QCVN 26:2010 daytime value, as a descriptive statistic and not as a compliance rate. Per-site exceedances from `results/tables/hanoi_exceedances.csv`. Mean level peaks at hour 18 (71.7 dB).

Four things in Figure 2 bear on everything that follows. The three areas are roughly 11 km apart and nothing between them was measured, which is the envelope the grid-extent test (`tests/test_grid_extent.py`) exists to enforce. Within each area the coverage follows the street network rather than a grid, so the sample is road-biased by construction, 189 of 363 measurements are roadside, from `dist_to_road`, and a model fitted on it inherits that bias. The areas are not equal: Ocean Park carries half the campaign. And **the video covers 39 % of it**, which is the ceiling on anything result 3 (§4.4) could ever have shown: the traffic term was fitted where the traffic was actually counted, and that is 142 points, not 363.

The capture app's burned-in location is wrong, and it matters for the next campaign

The videos carry a date and a place name burned into the corner by the capture application. The date is right. ***The place name is not.*** *It is cached from the last geocode the app made, so all 42 recordings taken in Vinh Tuy on 13 July are captioned* Ocean Park Gia Lam*, 8 km away, and the 14 July recordings in Hoan Kiem carry no place line at all. The GPS tag in the file metadata is correct throughout and agrees with the measurement coordinates; it is what* `scripts/index_videos.py` *reads and what places every video on Figure 2.*

Nothing here was sited from the caption, so nothing in §4 is affected. It is recorded because the planned campaign puts a camera in the hands of 60 volunteers: if they use this app, every video will carry a confident and sometimes wrong location, and whoever processes them afterwards will have no reason to doubt it. Read the GPS tag, never the caption.

4.2 Result 1, a three-parameter physical law beats every learned model

The delivered model treats each road class as an incoherent line source, whose intensity falls as $1/d$ rather than $1/d^2$:

$$E(x) = \frac{A_{\text{hw}}}{\max(d_{\text{hw}}, d_0)} + \frac{A_{\text{res}}}{\max(d_{\text{res}}, d_0)} + B, \qquad L(x) = 10 \log_{10} E(x)$$

with d_{hw} and d_{res} the distances to the nearest major axis and to the nearest local street, separated by OSM `highway` tag. Fitted values (`models/metrics.json`, `physical_params`): $A_{\text{hw}} = 4.774 \times 10^7$, $A_{\text{res}} = 3.800 \times 10^7$, $B = 1.106 \times 10^{-10}$, $d_0 = 5$ m; residual s.d. 6.07 dB.

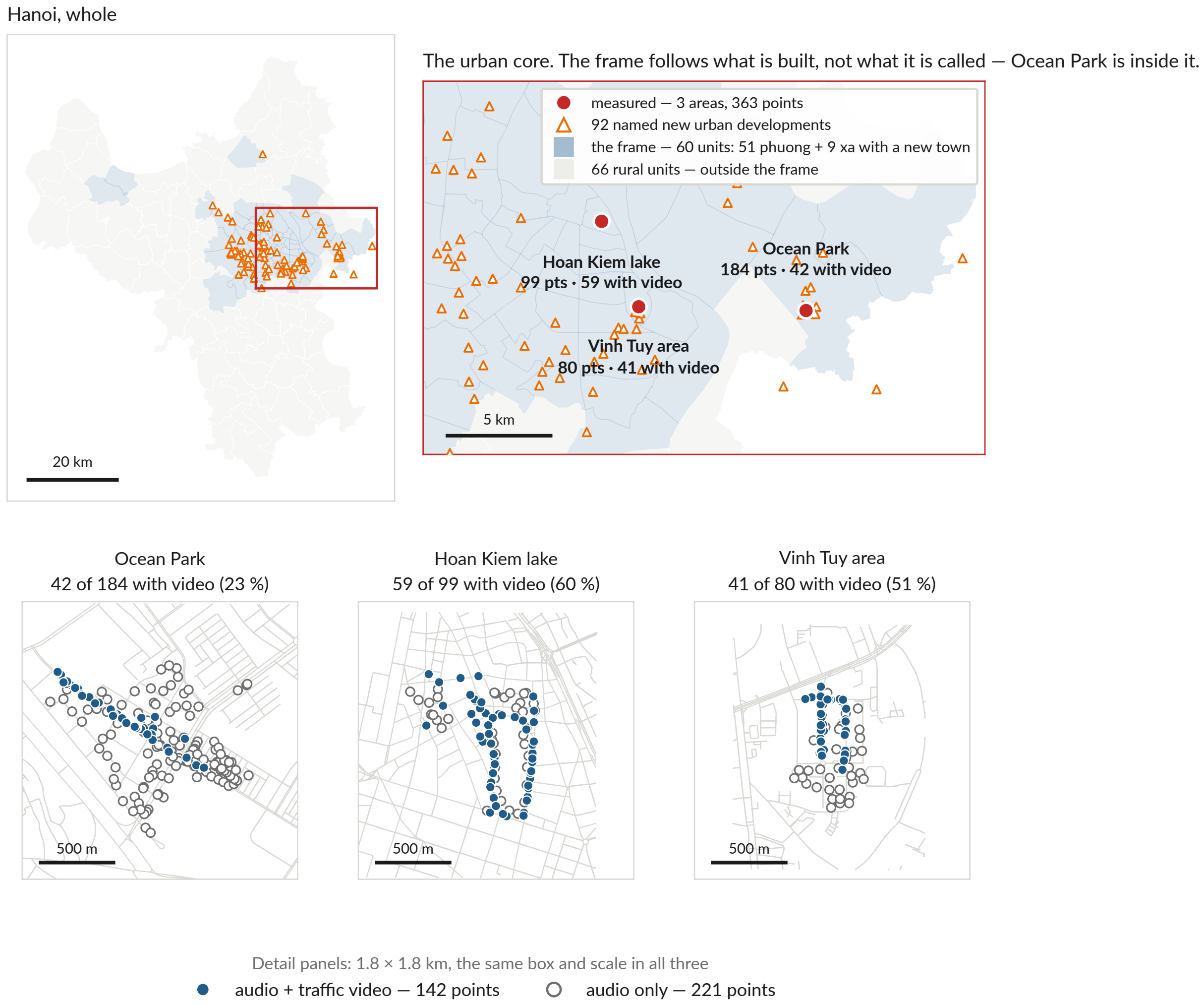
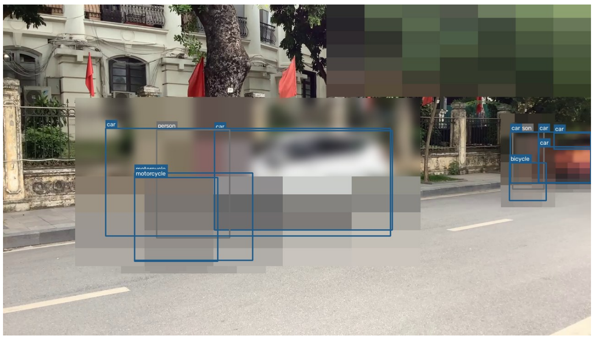


Figure 2: Where the 363 measurements are, and which of them carry video. **Top left:** Hanoi whole, with the locator box for the panel beside it. **Top right:** the urban core, the sampling frame of §6.2 in blue, the 92 named new urban developments as triangles, and the three measured areas. **Bottom:** each area cut to the same 1.8×1.8 km box, so the three densities can be compared by eye; panels scaled to their own data would make the sparsest site look the busiest. *The campaign runs two instruments and only one of them ran everywhere.* **142 of the 363 measurements have a traffic video matched to them and 221 do not**, and the shortfall is not spread evenly: Ocean Park is the largest site and the thinnest covered, at 23 % against 60 % at Hoan Kiem. The matched pairs are the pipeline's own (`data/processed/vehicle_counts.csv`), not recomputed here. Roads are the OSMnx extracts the GAMA simulation runs on; boundaries are `hanoi_wards.geojson`. `scripts/12_presentation_figures.py`.



Hoan Kiem, 14 July 17:26, old quarter

Vinh Tuy, 13 July 17:49, transport corridor

Ocean Park, 13 July 08:13, a new town still under construction

Figure 3: Three of the 147 traffic videos, one per typology, with the detector's boxes drawn on. **Every frame is redacted before it is written to disk** and no raw frame is kept: each is passed through the same `yolov8n` the counting pipeline uses, every *person* box is dilated and mosaicked, and every vehicle is blurred whole and mosaicked over its lower half where a plate sits (`scripts/make_video_stills.py`). The footage carries identifiable faces and readable plates, it is not distributed, and this repository is public. The redaction is a machine pass and is stated as one: a person the detector misses is a person the mosaic misses, which is why every still here was also looked at by a human before it was included. The blurred patch in the top right corner of each frame is the capture app's burned-in caption, removed for the reason given in §4.1.

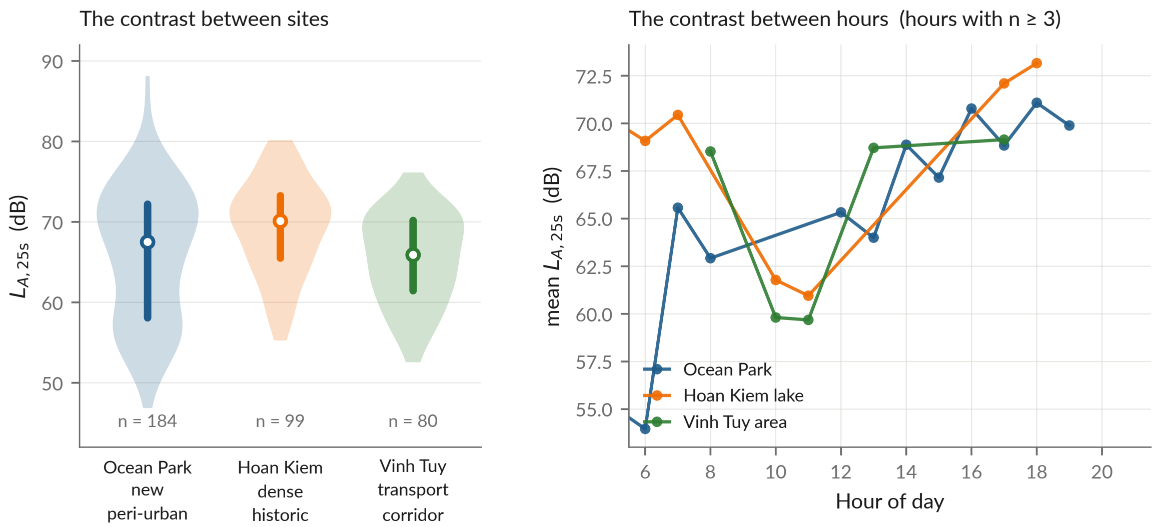


Figure 4: The 363 measurements: spatial layout, level distribution and temporal coverage. Produced by `scripts/12_presentation_figures.py` from `data/processed/measurements.csv` and `models/metrics.json`.

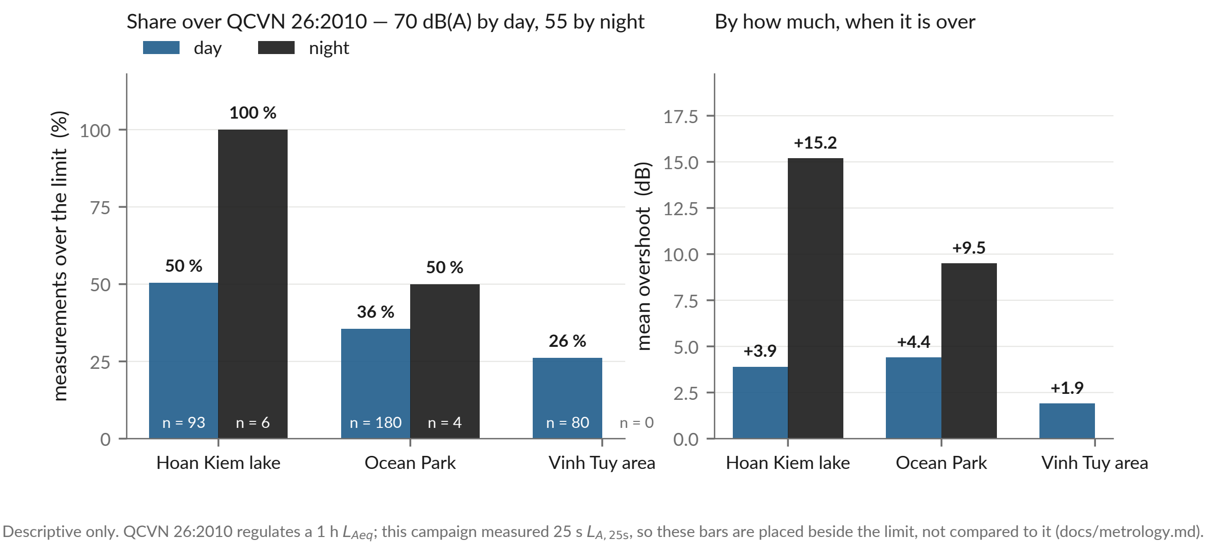


Figure 5: The same exceedances, drawn, with the sample size printed inside every bar, because the night bars are the ones a reader will quote and the ones the campaign cannot support. *Every* night measurement at Hoan Kiem is over the limit, and there are six of them; Vinh Tuy has none at all. Ten night measurements out of 363 is the whole night evidence of this campaign. Read from `results/tables/hanoi_exceedances.csv`.

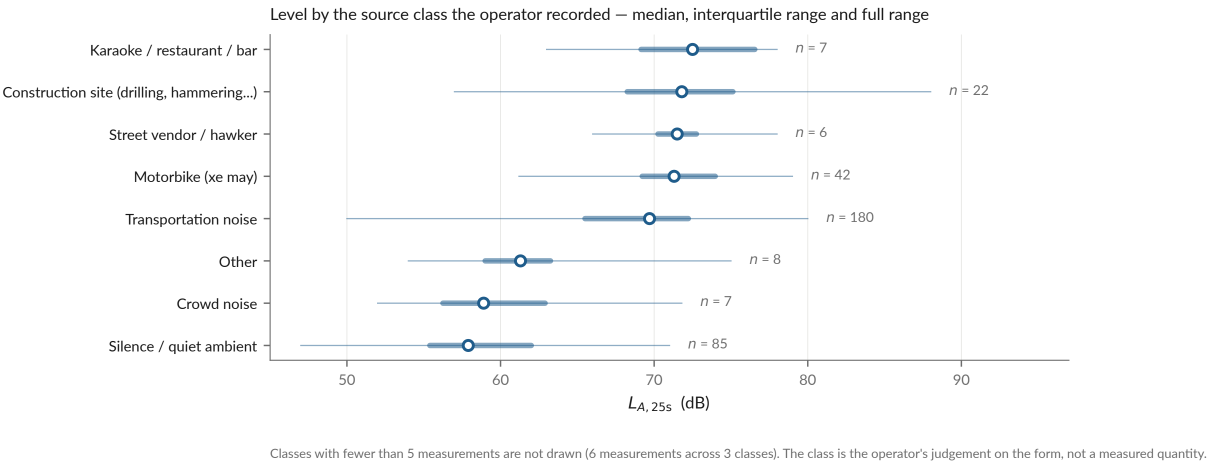


Figure 6: Level against the source class the operator selected on the form, the one variable in the dataset that is a human judgement rather than a sensor reading. It behaves: the ordering it produces is the ordering the physics predicts, karaoke and construction at the top, declared silence at the bottom, with no reordering needed. It also shows how unbalanced the campaign is, *transportation noise* and *silence* carry three quarters of it. Recomputed from `data/processed/measurements.csv`.

Model	Block-CV 600 m	Buffered LOO 300 m	Leave-one-site-out
Mean by (site, hour)	−0.008	−0.419	−0.058
Regression on $\log d_{\text{road}}$ (2 par.)	0.221 [0.09, 0.29]	0.200 [0.08, 0.27]	0.189 [0.06, 0.27]
Physical kernel (3 par.), delivered	0.255 [0.09, 0.36]	**0.246 [0.10, 0.34]**	**0.222 [0.05, 0.33]**
LightGBM v1 (6 features)	0.304 [0.10, 0.46]	0.137 [−0.10, 0.33]	0.029 [−0.35, 0.21]
LightGBM v2 (8 features)	0.332 [0.11, 0.49]	0.099 [−0.13, 0.25]	−0.035 [−0.40, 0.16]
Hybrid, physics + ML residual	**0.395 [0.18, 0.53]**	0.123 [−0.22, 0.29]	0.035 [−0.41, 0.23]
Hybrid, capacity-restricted residual	0.378 [0.20, 0.51]	0.144 [−0.15, 0.35]	0.106 [−0.27, 0.27]

R^2 with 95 % block-bootstrap intervals, $n = 363$, 17 blocks, 2 000 replicates. All values from `models/metrics.json` via `models/model_comparison.md`. Best per column in bold.

The point of the table is the inversion. Read the first column and the hybrid leads; read the third and it is nearly worthless. The ranking reverses monotonically as the split begins to test generalisation, the signature of capacity fitting spatial autocorrelation rather than physics. Three further readings hold:

- **The 300 m morphological aggregates have negative marginal value.** Adding built-area ratio, road density and intersection count to distance-to-road *degrades* prediction, by 0.07 R^2 under block-CV, 0.16 under buffered LOO and 0.18 under leave-one-site-out.
- **What remains of the ML model is time, not morphology.** The gap between the full model and the morphology-only ablation under block-CV (0.304 vs 0.153) is carried by the hour features. Under leave-one-site-out the time-only ablation is itself negative (−0.139).
- **The hybrid was built, tested and rejected**, not deferred. Against the bare physical core it gains $\Delta R^2 = +0.140$ under block-CV and loses −0.123 and −0.187 under the two strict protocols. The learned residual is computed at every run and deliberately not applied (`apply_residual: false`).

4.3 Result 2, cross-city transfer fails, in three distinct ways

Re-run for this report on **24 August 2026** with `scripts/experiments/uganda_transfer.py`. It needs no Ugandan data: the two Uganda boosters are versioned in `models/`, and it evaluates them against Hanoi's own 363 points.

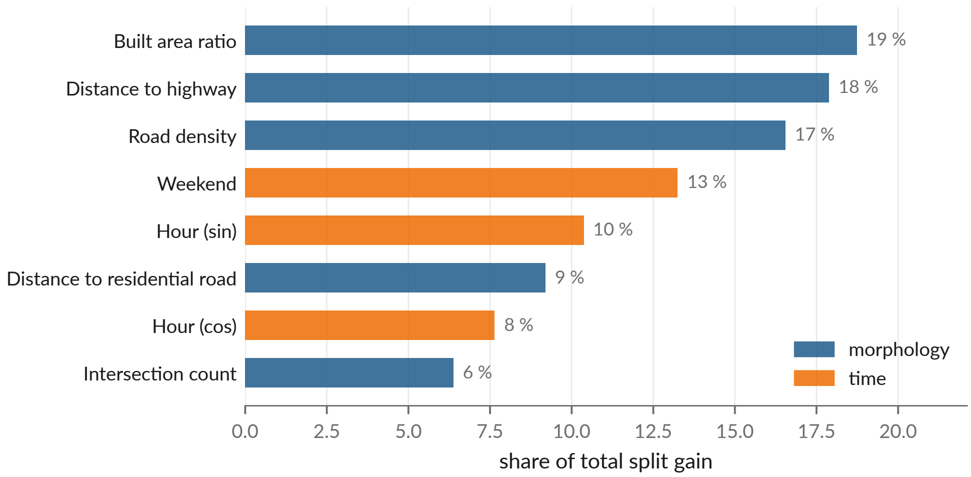


Figure 7: Gain-based feature importance inside the fitted LightGBM, kept because it explains the ablation: distance to road is the single largest contributor and matches the three area-aggregated descriptors combined. `scripts/12_presentation_figures.py`.

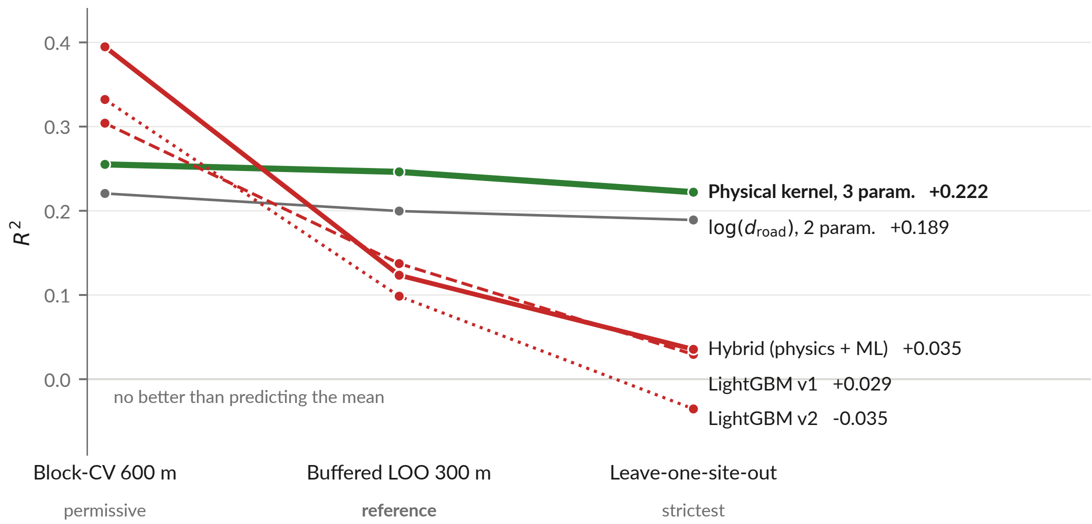


Figure 8: The ranking inversion, drawn. Same models, same folds, three protocols of increasing severity. `scripts/12_presentation_figures.py`.

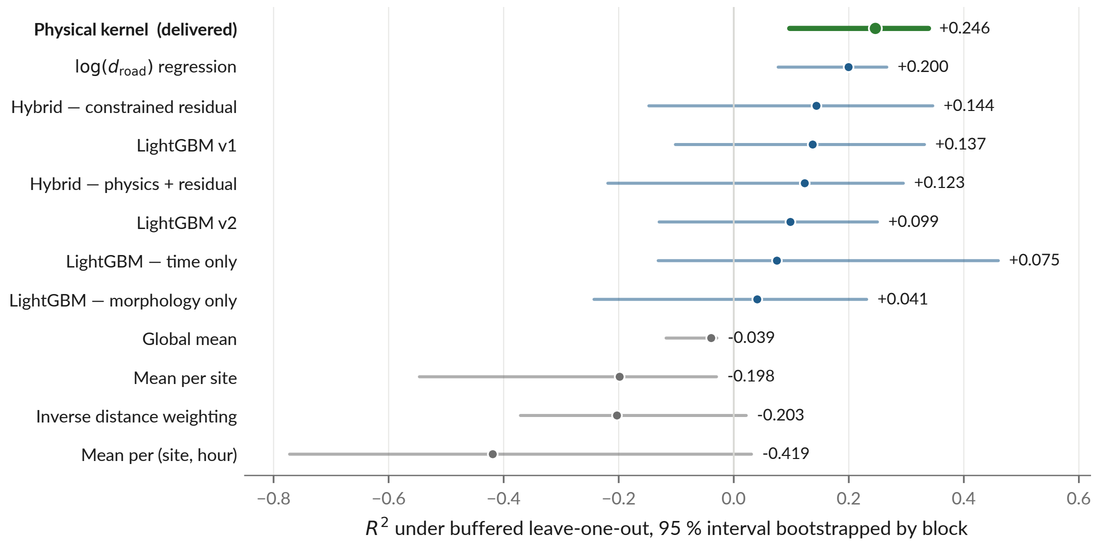


Figure 9: Every model under the reference protocol with its 95 % interval. The physical kernel is the only model whose interval excludes zero under all three protocols, and its interval overlaps that of the two-parameter distance regression almost entirely (§5.3).

Pretrained model	**as delivered**	**mean diff. removed**	**and rescaled ($=r^2$)**	r
Uganda 61K (v1)	$R^2 = -15.86$	-2.14	$+0.141$	$-\mathbf{0.376}$
Uganda convention-invariant (v2)	$R^2 = -8.17$	-0.95	$+0.006$	$+0.076$
For scale, fitted locally on the same 363 points:				
log(distance to road) alone	$R^2 = +0.240$ (in sample)			
Physical kernel, buffered LOO	$R^2 = +0.246$			

- **It is not a calibration offset.** The Kampala model predicts about 26 dB below Hanoi (MAE 26.5 dB, bias −26.4 dB), and removing that difference still leaves R^2 negative. A recalibrated transfer would not work either.
- **v1 transfers anti-information.** Its correlation with Hanoi is *negative*: it ranks quiet and loud the wrong way round. The only way to extract anything from it is to flip it.
- **v2 removes the mapping-convention artefact and leaves nothing.** $r = +0.076$; even granted a free bias and a free slope its ceiling is $R^2 = 0.006$. Not wrong, empty.

One locally fitted distance term carries more than a 61 000-point model trained in another city carries at all.

A small drift against the August brief

`docs/presentation-brief.md` §2.3 records $-15.8/-2.17/+0.151/r = -0.388$ for v1 and $-8.1/-0.96/+0.004/r = +0.066$ for v2. Today's rerun gives the values tabulated above. The differences are small but not pure rounding: either the brief was computed at an earlier commit, or the experiment has a non-deterministic component we have not isolated. The manuscript must quote one run and name it.

4.4 Result 3, vehicle counts from video carry no recoverable acoustic signal

147 videos were matched to measurements (median offset 15 s, p90 56 s, max 161 s). A non-negative least-squares regression in *energy*, $E_{\text{total}} = E_{\text{background}(s)} + \sum_c n_c e_c$ with $e_c \geq 0$, returns **exactly zero** for motorcycles and cars, in both passes: per-frame density first, then, after implementing the project's own recommendation, ByteTrack line-crossing flow.

Pooled over the 147 matched videos, total flow against measured level gives $r = -0.109$ and motorcycle flow $r = -0.004$ (recomputed from `data/processed/vehicle_counts.csv`). The one structured exception is Vinh Tuy, the transport corridor, the only site with a positive association, which *strengthens* when density is replaced by flow ($r: +0.22 \to +0.30$).

The diagnosis in `docs/negative-results.md` §5.x is that the failure is a mismatch between the quantity measured and the quantity that governs the physics: density is a stock and emission is driven by flow; speed dominates above 30–40 km/h and is unobservable in a frame count; parked vehicles are counted; and distance is discarded because the field of view is not georeferenced.

This result should be reframed before submission

The August data audit (recorded in `presentation/main.tex`, slide "Findings 2 and 3") found the detector's modal split to be ***57 % cars and 36 % motorcycles*** *by density, reproduced exactly from `vehicle_counts.csv` for this report, in a city where reality is roughly the inverse. The matched subset also spans 61–80 dB against 47–88 dB overall, cutting the standard deviation by 42 % (4.12 against 7.15 dB). A negative correlation between traffic and noise is not physical. The audit's own conclusion is that this is far more likely an* ***instrument failure than a finding about acoustics****, and that it should be reported as "the chain could not be made to work".* The current wording in `docs/negative-results.md` has not yet been changed to match.

4.5 How good can any spatial model be here?

$R^2 \approx 0.25$ invites the question, so the project answers it with a ceiling. Two measurements taken close together *at the same hour* must be predicted identically by any spatial model, so whatever they disagree by is variance no spatial model can explain. Recomputed for this report from `measurements.csv`:

Separation	Pairs	Mean |difference|	Implied σ	Ceiling on R^2
< 20 m	29	4.23 dB	3.75 dB	0.72
< 40 m	87	5.44 dB	4.82 dB	**0.54**
< 60 m	229	6.09 dB	5.40 dB	0.43

Conversion: for two readings of a common value with error s.d. σ, the difference is $N(0, 2\sigma^2)$, so $\sigma = \overline{|\Delta|}\sqrt{\pi}/2$. Total variance of the 363 measurements is 51 dB^2.

At the 40 m scale the map works at, the ceiling is around $R^2 \approx 0.54$. The delivered model reaches 0.246, about half of what is attainable, not a tenth of it. **The caveat belongs on the same page**: pairs at the same hour may fall on different days, so this σ carries day-to-day variation and overestimates pure measurement noise; and 29 pairs is few. It is an order of magnitude, not a bound.

4.6 The published map and the simulation

The exported grid is 5 587 cells at 40 m over three zones × 17 hours (`results/maps/hanoi_noise_map.csv`), and it covers the sampled envelope only, the three measured sites plus a 400 m margin.

This validation was itself a lesson

An earlier published validation reported MAE 3.79 dB. It had validated a grid that was regenerated after it, and the drift was found on 11 August by re-running the script during an unrelated task, not by any check. The flattering figures were the stale ones. The old artefact is archived with its reason (`docs/archive/validation-2026-08-05/`) and the Makefile now declares the dependency so `make` rebuilds it.

4.7 What could not be regenerated

Eight figures in `results/figures/sunbird/`, the Uganda reproduction, are frozen images that never had a scripted producer: they come from notebook cells whose input data is absent from the repository. They are marked as such (`NOT-REGENERABLE.md`) rather than quietly reused. One

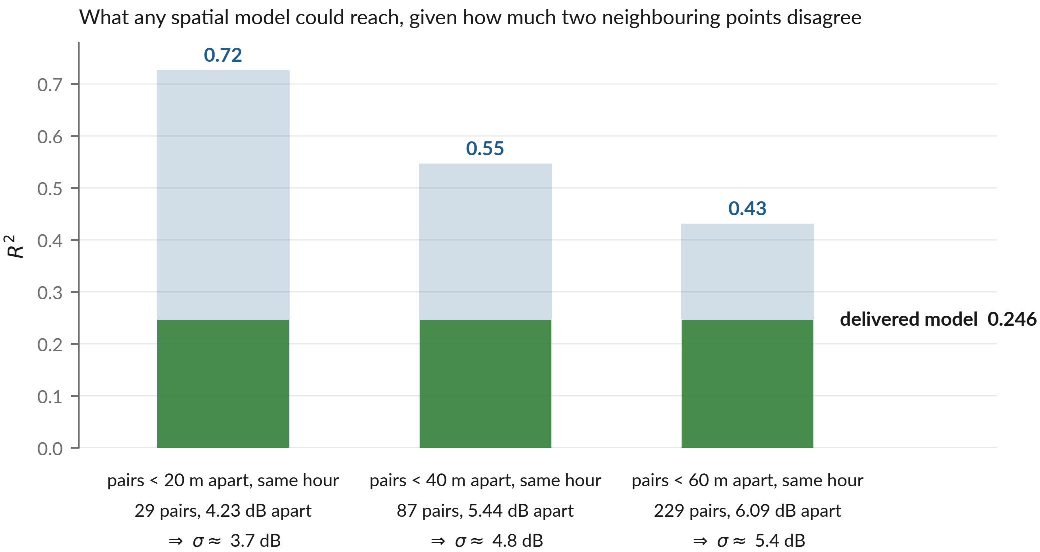


Figure 10: The ceiling, drawn. `scripts/12_presentation_figures.py`.

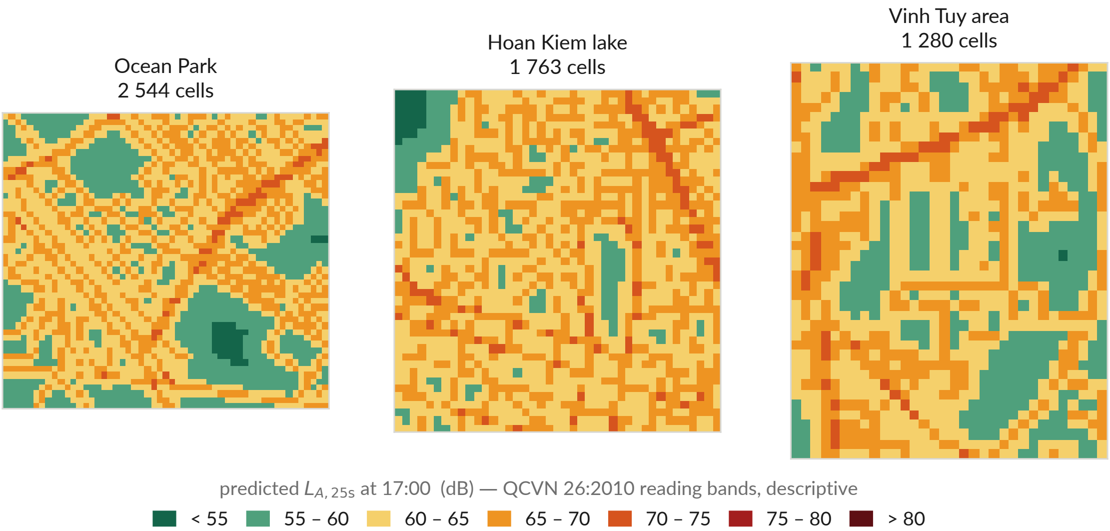


Figure 11: The published 40 m grid, redrawn from the CSV rather than screenshotted. QCVN 26:2010 colour bands, used descriptively.

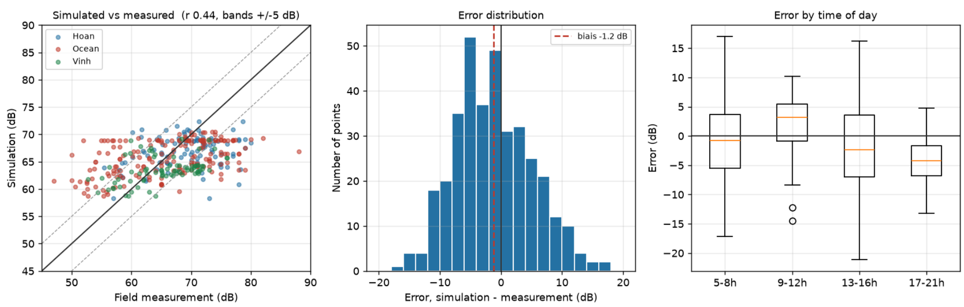


Figure 12: Simulated against measured levels, $n = 360$ matched points (`results/tables/validation_simulation.csv`, `scripts/08_validate_simulation.py`). Mean error -1.24 dB, MAE 5.30 dB, RMSE 6.49 dB. The simulation reproduces the spatial and temporal ordering; it does not reproduce the spread, which is expected given that features aggregated over 300 m cannot resolve the facade/courtyard contrast.

of them, `pred_vs_real.png`, carries "$R^2 = 0.250$" for the *Uganda* reproduction, which is easily mistaken for the Hanoi delivered model's 0.246; the two are unrelated. **No such figure is used in this report.**

5 Discussion, limitations and open questions

5.1 The two limits that are not fixable by code

No absolute reference. Three handsets were cross-calibrated against each other and against no standard, so every level in this project is relative. `docs/metrology.md` argues this well, but a reviewer in acoustics will ask whether one afternoon beside a class 1 or class 2 meter was truly out of reach. It would convert "calibrated in relative terms" into "calibrated" and would open the comparisons the work currently forbids itself.

Three sites are three typologies. The effective number of independent morphological configurations is close to three whatever the number of points, and three is not a basis for claiming coverage of a city. This bounds generalisation further than the R^2 does, and it is the reason every published map stops at the sampled envelope.

5.2 The open finding: a -5 dB discontinuity on 27 June

This is, in our assessment, the most serious unresolved item, and it is **not yet written up anywhere in `docs/`**, it exists only in the August deck (`presentation/main.tex`, frame "Finding 1") and in `scripts/12_presentation_figures.py`.

Verified from `measurements.csv` for this report: the mean level before 27 June is 69.75 dB ($n = 148$) and after it 64.66 dB ($n = 215$), a raw gap of -5.09 dB. On the same date the share of integer-valued readings falls from **87 % to 20 %**, which is the signature of a change in the recording instrument or form rather than of the city getting quieter.

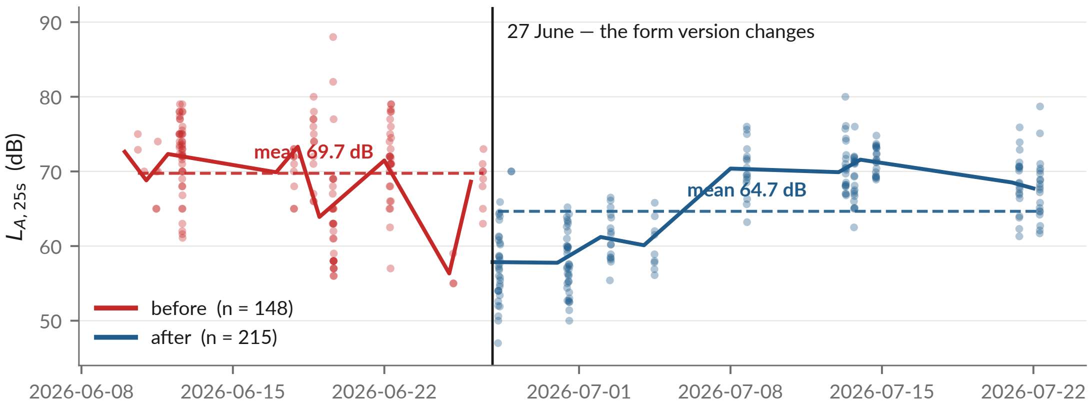


Figure 13: The level distribution around 27 June 2026. The break date is not chosen by the figure: it is the date the form version changes. Produced by `scripts/12_presentation_figures.py` from `data/processed/measurements.csv`.

The deck reports a controlled effect of -5.14 dB (SE 0.55, $p \approx 10^{-20}$) and states that before/after is predictable from (lat, lon) alone at AUC 0.83, so an instrumental shift is *spatially structured*, and a model can learn it as morphology.

What we could and could not reproduce

The raw gap and the integer-reading collapse reproduce exactly. The controlled -5.14 dB *does not: no script in the repository computes it, so the specification behind it (which covariates, which encoding) cannot be recovered. Refitting with site, class and hour dummies gives* -7.75 dB *(SE 0.71), same sign, same order, different number. Until the specification behind the* -5.14 dB *figure exists as code in the repository, neither figure should go in the manuscript.*

The buffered leave-one-out protocol excludes 300 m around each test point. **It does not exclude a period.** That is the gap, and the indicated fix is to diagnose the cause and refit with a phase term.

5.3 The headline ranking is not statistically separated

Physical kernel $R^2 = 0.246$, CI $[0.097, 0.339]$, against $\log d_{\text{road}}$ at 0.200, CI $[0.077, 0.266]$. The intervals overlap almost entirely. "The physical kernel beats log-distance" is, on this sample, a coin flip. The stronger and better-supported claim is that *both simple geometric models beat every learned model under every protocol that tests generalisation, and they are the only models that barely move between protocols.* Every summary table in the manuscript must carry the intervals.

5.4 Other known weaknesses

- **The vehicle detector has never been validated** against manual counts. Precision, recall and MAPE per class are unknown, and two-wheeler under-detection is expected but unquantified. *Modal shares must not be published until it is.* One day of manual counting on ten videos turns an open-ended limitation into a quantified uncertainty.
- **The Uganda chain (notebooks 01–06) is not reproducible** on a fresh machine: it expects `data/processed/uganda/*`, which exists nowhere. The transfer *experiment* is now reproducible from the versioned boosters, but the pretraining is not.
- **Night is not sampled.** 10 measurements fall outside 06:00–21:00 and there are none between 22:00 and 03:00.
- **15 % of `construction_nearby` is inferred** from the noise class rather than observed, and the published CSV does not flag observed against derived values.
- **Technical debt**: `04_evaluate_models.py` is 598 lines holding three CV protocols, the bootstrap and six model definitions, the most valuable code in the project is the least testable; `07_export_gama_inputs.py` is 414 lines doing five jobs.

5.5 Inconsistencies found while writing this report

Small, but each one contradicts the project's own single-source rule, and each is a five-minute fix:

1. **The test suite does not currently pass.** In `tests/test_pipeline_end_to_end.py`, `test_every_pipeline_step_is_present` asserts scripts `01`–`11`, and script `12` was added on 21 August. Actual state on 24 August 2026: **41 passed, 1 failed, 1 deselected**.
2. **The README badge says "34 tests"**; the suite collects 42.
3. **The README's night claim is wrong in both halves.** It says "10 measurements after 21:00, none 00:00–05:00". In fact one measurement is at hour 21 and nine are at hours 4–5. The count of 10 is right; the window is not.
4. **One exceedance figure lives in two places and they disagree**: `docs/literature-review.md` says 38.6 %, `results/report/numbers.tex` says 37.4 %. The data gives 37.4 % for daytime samples above 70 dB.
5. **The transfer numbers in the presentation brief do not match today's rerun** (§4.3).

6. **The August data-audit findings are not in `docs/`.** The deck cites `docs/audit/` for the 27 June discontinuity; that directory contains the 5 August scientific audit and the restructuring documents, not this one.

6 Planned deployment of the field application

Everything above is what we built and what it does or fails to do. This section is the one forward-looking part of the report, and it is separated deliberately: none of it is a result. It is the plan the application was built for, and the negative results of §4 are what constrain it, each of them being a limit the design below has to respect.

6.1 Handover: the APK goes to VinUniversity

The application (§3.7, Figure 1) leaves the team as a signed APK handed to **Nguyen Thanh Quang** at VinUniversity, who distributes it inside the university and recruits from there. We are not the distributors: neither of us is in Hanoi past August, and an application that collects strangers' positions should be handed out by someone with an institutional relationship to the people installing it, not by a departing intern over a messaging app.

Three things must be true before the APK is passed on, and only the first is technical:

- **The KoboToolbox account behind the outbox must be institutional**, not a student's. Whoever owns it holds participants' GPS positions and timestamps. This is item 2 of §7.1, and it blocks distribution, not just publication.
- **A VinUniversity ethics review must exist** before any collection beyond the team itself. The video collection of June–July went without one; the repository says so, and that must not be repeated at a scale with volunteers and GPS traces.
- **The consent screen must state what the application uploads**: the audio clip, the coordinates, the timestamp, and what it does not.

The application is honest about its own limits and should stay that way in the hands of volunteers: it refuses to predict outside the three mapped areas, it makes no compliance claim, and its calibration screen has never been used against a reference instrument. The campaign below is what would finally use that screen.

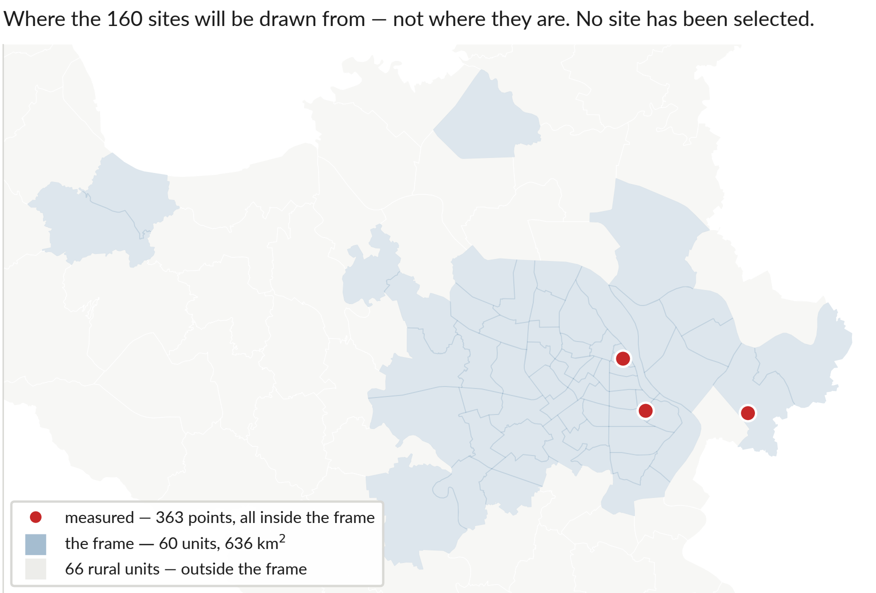
6.2 The campaign the application is being handed over for

The design is a noise-measurement campaign in Hanoi with two arms, a *mapping* arm and an *individual exposure* arm, run in partnership with a Vietnamese institution. It is recorded here as specified; none of it has been executed, and no number in §4 depends on it.

Drawing the frame found a problem in it, and the frame was widened

Placing the three measured areas on the ward boundaries was meant to be illustration, and returned a finding instead. Under the frame as first specified, the 51 urban phuong*, **Ocean Park was not in it**. All 184 of its measurements, half this campaign, fall inside* Xa Gia Lam*, which is administratively a rural commune and in fact a new town of tower blocks, a mall and a construction site (Figure 3, bottom).*

***The frame has been widened, and the criterion is what is built rather than what it is called.** A unit is in the frame if it is a* phuong*, or if it contains a named new urban development,* khu do thi *and the developer names that do not use the term. That is computed from OpenStreetMap, not listed by hand: 92 such developments across Hanoi, 89 of them inside an administrative unit, admitting **9** **xa** alongside the 51* phuong*. The frame is **60 units and 636 km^2** rather than 51 and 407, and Ocean Park is inside it, as are An Khanh, Hoai Duc, Thanh Tri, Dong Anh and the rest of the new-town belt. A noise study of a growing city that sampled only the units the map still calls urban would have missed the growth.*



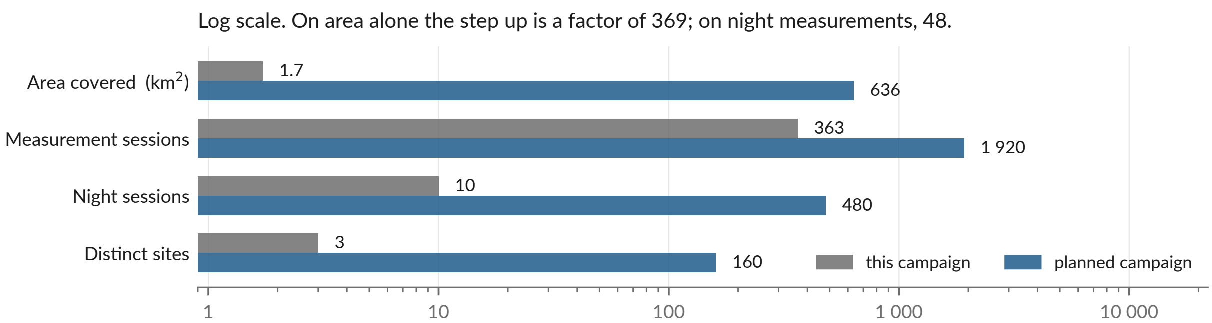


Figure 14: The sampling frame, and the size of the step. **Nothing on this map is a site**: the 160 sites have not been drawn, and 160 invented coordinates would commit the team to a sample nobody selected. What is drawn is the frame: the 51 urban *phuong* of Hanoi under the July 2025 reform plus the 9 *xa* that contain a new urban development, 60 units and 636 km^2, against the 1.7 km^2 the campaign has actually covered, a factor of 369. Boundaries are OpenStreetMap (`data/processed/hanoi_wards.geojson`, fetched by `scripts/fetch_hanoi_wards.py`); areas are computed in UTM 48 N, not from degrees. `scripts/12_presentation_figures.py`.

Mapping arm, 160 sites, 1 920 sessions. 160 sites are drawn in **8 strata of 20 sites** each: major axes and ring roads; secondary streets; the old quarter and its narrow streets; residential alleys (*ngo*); markets; sensitive receivers (schools and hospitals); quiet zones (lakes and parks); and construction and industrial sites, to which the widening of the frame adds a ninth in all but name, since the new towns now inside it are a typology of their own: wide new roads, low current density and live construction, which is what Ocean Park already showed. The draw is made across the **60 units** of the frame (Figure 14): the 51 urban *phuong* of Hanoi under the division in force since July 2025, plus the 9 *xa* that contain a new urban development.

Each site is measured as L_{Aeq} **over 15 minutes** on four time windows, 07:00–09:00, 11:00–13:00, 17:00–19:00 and 22:00–24:00, repeated on **two working days and one weekend day**. That is $160 \times 4 \times 3 = \mathbf{1\,920}$ measurement sessions.

The field force is **60 active volunteers**, recruited at **90** to absorb the 30 % attrition that citizen-science campaigns of this length see. They are organised in **15 teams of 4**, each team covering 11 sites, with **night work in pairs, without exception**, under **3 coordinators**.

Individual exposure arm, a stratified panel. A panel stratified by neighbourhood type and travel mode, targeting **400 active participants** (50 per stratum), recruited at **1 500** for the same attrition reason.

Instrument and protocol. Measurements are taken with the **NoiseCapture** application on a **whitelist of 5 to 8 handset models**, iOS preferred. Each model carries **its own correction curve**, established against a class 1 or class 2 sound level meter at 50, 60, 70, 80 and 90 dB(A), with **automatic gain control disabled**; the curve is re-verified **monthly**, and at the start and at the end of the campaign.

In the field: the handset is held **at arm's length, 1.2–1.5 m above the ground and more than 2 m from any reflecting façade**. **No measurement in rain, and none in wind above 5 m s^{-1}.** Alongside each session, the operator records **two-wheeler and car flow and a horn count**, and L_{A10} **and** $L_{A\max}$ **are computed in addition to** L_{Aeq}.

The video arm, and why it is not optional this time. Volunteers record a **traffic video at the same time as the level**, so that flow is measured rather than estimated. This campaign already did that and did it unevenly, 142 of 363 measurements carry video, 23 % of them at Ocean Park (Figure 2), and result 3 was fitted on that thin, uneven half. Three things follow for the plan, and all three are cheap:

- **Fix the video share and state it up front.** Whether every session carries video or a defined sub-sample does, it must be a decision rather than a by-product of who remembered. A stratum with 20 % video coverage cannot carry a traffic term.
- **Record at the resolution the detector will be run at.** The counting pipeline currently runs at `imgsz=640` and finds 30 % fewer vehicles than the same frames at 1280 (§6.2 above, §7.2). Fix the inference resolution before the campaign, not after.
- **Trust the GPS tag, never the burned-in caption** (§4.1). With 60 volunteers this stops being a curiosity and becomes a systematic mis-siting of the video record.

The footage is personal data the moment it exists: it carries faces and plates, and 6 GB of it already has no custodian (§7.1.1). Any campaign that multiplies it by ten needs the retention decision made *first*, and a redaction step in the pipeline rather than at publication time, the one in `scripts/make_video_stills.py` is a start and is only good enough for three hand-checked stills.

Reference and governance. Results are read against **QCVN 26:2010/BTNMT**: 70 dB(A) from 06:00 to 21:00 and 55 dB(A) from 21:00 to 06:00 in ordinary zones, 55/45 dB(A) in sensitive zones. The campaign runs **in partnership with a Vietnamese institution**, VNU University of Science, the Hanoi University of Civil Engineering, or the VAST Institute of Environmental Technology, with a **research permit**, a **consent protocol**, and an **ethics opinion covering participants' GPS traces**.

What this campaign inherits from §4

The protocol above answers, point by point, the three things that broke here. It fixes an instrument per handset model against a reference meter, which is the absolute-scale anchor this project never had. It measures L_{Aeq} over 15 minutes rather than 25 seconds, which is the quantity QCVN 26:2010 actually regulates, so its readings, unlike ours, may be compared to the standard rather than merely placed beside it. And it stratifies the draw over the whole city instead of three sites, which is what would let a spatial model be evaluated outside the envelope it was fitted in. Two arms, one instrument, one correction curve per model: the failure modes of §4.2–§4.4 are each addressed by a specific line of it.

7 Next steps and timeline

Dates are the repository's where it has them.

7.1 Blocking before any public release

1. **Video retention decision**, custodian, storage location, deletion or anonymisation deadline. 6 GB of unanonymised video showing faces and plates currently sits with no owner and no expiry. This is the most consequential open item in the project. *Holder: supervisor.*
2. **Ownership of the KoboToolbox account** if the app is used for a public campaign, it should be institutional, not a student's, because whoever owns it holds strangers' GPS positions and timestamps. And **VinUniversity ethics review** before any new collection: the video collection went without one and the repository says so; that must not be repeated at larger scale.

7.2 Scientific work, in value-per-effort order

1. **Validate the YOLO detector** on ∼10 videos, double-counted manually, about one day. It is the input uncertainty of every downstream number and it unblocks the modal shares. There is now a reason to think the uncertainty is large and one-sided: `02_count_vehicles.py` runs inference at `imgsz=640` on 1280×720 footage, and re-running the same frames at `imgsz=1280` with the confidence threshold held at 0.30 finds **30 % more vehicles**, 78 against 60 across a 12-frame sample, more in 8 of the 12. Twelve frames is a hint, not a measurement, and the sign is what matters: the counts result 3 was fitted against are more likely to be undercounts than overcounts, and an instrument that undercounts by a variable amount is exactly what would flatten a real relationship into the zero coefficients of §4.4. Vary the resolution first; it costs an afternoon and it may reframe the whole negative result.
2. **Diagnose the 27 June discontinuity** and refit with a phase term, and commit the code that computes the controlled effect.
3. **Reframe negative result 3** as an instrument failure in `docs/negative-results.md` and in the manuscript.
4. **Carry the confidence intervals into every summary table.**
5. **Implement a real propagation kernel**, CNOSSOS-EU via NoiseModelling (Kephalopoulos et al., 2014; Bocher et al., 2019). This is the scientifically indicated continuation, not an optional extra: if a two-parameter distance term already beats a six-variable gradient-boosting model, then physical propagation corrected by a locally learned residual is the architecture the data points at. It was out of budget for a three-month internship; implementing it is months, citing what is missing is a page.

6. **Repeat a short campaign with a reference instrument** to anchor the absolute scale. The app's calibration screen exists for exactly this.

7.3 Deliverable track

Deliverable	State	Next action
Manuscript (Overleaf)	Two sections drafted in the repository first (`metrology.md`, `negative-results.md`); Overleaf is the source of truth	Reframe result 3; add the intervals; add the CNOSSOS future-work paragraph with Figure 2 of Bocher et al. (2019).
Capture / demo with the app	Working end to end; the three-gesture demo is scripted in `docs/presentation-brief.md` §5	Record it. Steps 1–2 (measurement, map) need no network; step 3 (GAMA) needs `gama-headless.sh` `socket 6868` reachable on the wifi address, not localhost.
AAMAS submission or presentation	Intended; the venue, track and deadline are the supervisor's to fix	The agent-based exposure model (`simulation/gama/`) is the natural hook; tier 2 pedestrian agents are the natural addition.
Decks	35-slide and 23-slide versions compiled, plus speaker scripts and a formula annex (`presentation/`)	Ready.
Field application, distributed	APK built and working offline end to end	Hand the signed APK to Nguyen Thanh Quang for distribution inside VinUniversity (§6.1), then open recruitment for the campaign of §6.2.

7.4 Deliberately abandoned

LSTM / ST-GNN benchmarking on Barcelona (those models need continuous time series this campaign does not produce; `barcelona_transfer.py` is kept as a documented dead end); emitter agents in GAMA; demolition audio; per-class vehicle emission coefficients. **Acoustic event localisation**, the "gunshot" direction, is a different project rather than a continuation: time-difference- of-arrival localisation needs clocks agreeing to about 35 ms for a ±12 m fix, continuous listening, and at least four sensors at known fixed positions. This campaign has 363 spot measurements from three unsynchronised handsets. It is strong as future work and wrong as a promise attached to this result.

A Provenance of every figure and recomputed number

Artefact	Producer
campaign.pdf, ranking-inversion.pdf, forest-bloo.pdf, ceiling.pdf, map-grid.pdf, discontinuity.pdf, feature-importance.pdf, hanoi-sites.pdf, exceedance.pdf, class-levels.pdf, campaign-frame.pdf	scripts/12_presentation_figures.py, reading models/metrics.json, data/processed/measurements.csv and results/maps/hanoi_noise_map.csv; hanoi-sites.pdf additionally reads the road shapefiles under simulation/gama/inputs/ and exceedance.pdf reads results/tables/hanoi_exceedances.csv, and campaign-frame.pdf reads data/processed/hanoi_wards.geojson. Re-run on 28 August 2026. The only change against the committed versions is the map's colour ramp: the old one ran green – yellow-green – lime – yellow and its three middle bands, which cover most of the mapped area, sat at the same lightness and read as one band. The ramp is now ordered by lightness, and every label that was set in green is now set in ink, green is a line colour in these figures, never a font colour. No number moved.
Application screens (Figure 1)	appscreens.tex, drawn from mobile/app/src/main/java/org/noisehanoi/mobile/ui/. Not device captures, see the caption.
validation_simulation.png	scripts/08_validate_simulation.py
hanoi_wards.geojson, hanoi_new_developments.geojson	scripts/fetch_hanoi_wards.py, two Overpass queries against OpenStreetMap, fetched 28 August 2026 and committed. 126 administrative units (51 *phuong*, 75 *xa*) and 92 named new urban developments. It is the only script in the project that touches the network, which is why it is unnumbered and outside the pipeline.
data/processed/video_index.csv	scripts/index_videos.py, reading the QuickTime GPS and creation-time tags of the 147 files in data/raw/videos/. 118 carry a coordinate. No image data. Unnumbered: a clone has no footage to run it on.
Figure 3, the three stills	scripts/make_video_stills.py, redaction by yolov8n, then checked by eye. No raw frame is written to disk.
Which measurements carry video (142 of 363)	joined in fig_sites_map on matched_timestamp in data/processed/vehicle_counts.csv, the pipeline's own match
The frame: 60 units, 636 km^2	frame_units in scripts/12_presentation_figures.py, every *phuong*, plus every *xa* containing a development from hanoi_new_developments.geojson
The 30 % detector-resolution gap (§7.2)	12 frames sampled at random_state=7, yolov8n at imgsz 640 against 1280, conf held at 0.30. A hint at this sample size, not a measurement.
Frame area, measured area, the 369× ratio, and which areas fall inside the frame	computed in fig_campaign_frame from hanoi_wards.geojson and measurements.csv, reprojected to UTM 48 N
Model table (§4.2)	models/metrics.json via models/model_comparison.md
Transfer table (§4.3)	scripts/experiments/uganda_transfer.py, re-run 24 August 2026
Site table, exceedances, hour peak, night counts	recomputed from data/processed/measurements.csv and results/tables/hanoi_exceedances.csv
Ceiling table (§4.5)	recomputed from data/processed/measurements.csv
Modal split, match gaps, flow correlations	recomputed from data/processed/vehicle_counts.csv
Discontinuity raw gap and integer-reading share	recomputed from data/processed/measurements.csv
Simulation validation statistics	recomputed from results/tables/validation_simulation.csv ($n = 360$)
App statistics	counted in mobile/app/src/
Test-suite state	pytest tests -q, 28 August 2026

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